

Overview

The TP4054 is a single-cell lithium-ion battery constant current/constant voltage linear charger. Its simple external circuitry makes it ideal for portable device applications, suitable for both USB power and adapter power supply. It features internal reverse charging protection, eliminating the need for an external isolation diode. Thermal feedback automatically adjusts the charging current to limit chip temperature under high-power operation or high ambient temperatures.

The TP4054 has a charging cutoff voltage of 4.2V, and the charging current can be set via an external resistor. The TP4054 automatically terminates the charging process when the charging current drops to 1/10 of the set value.

When the input voltage is removed, the TP4054 automatically enters a low-current standby state, reducing the standby current to 3uA.

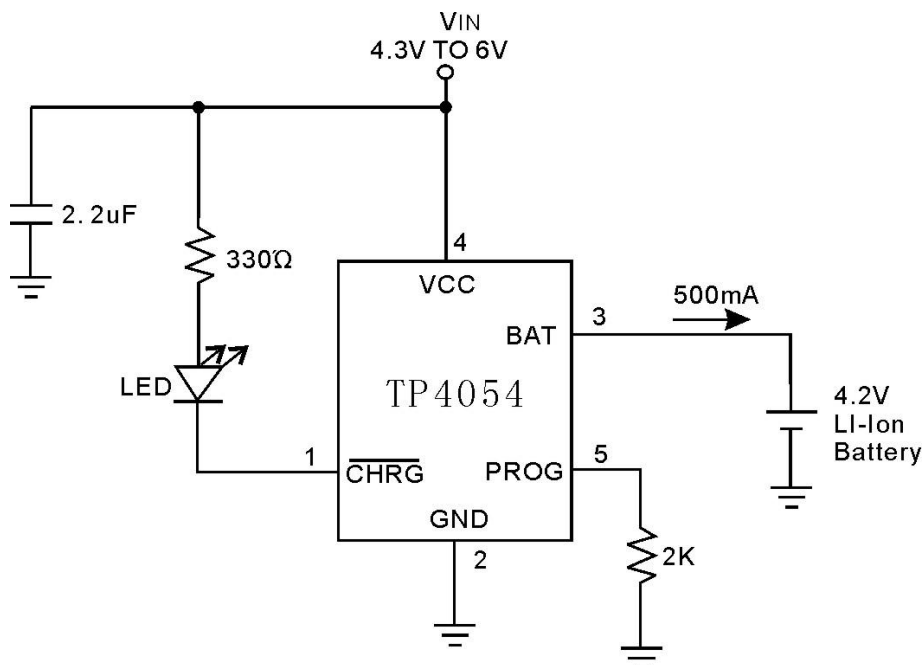
Features

- Maximum charging current: 600mA
- Flashing indicator light for poor battery contact
- No MOSFET, sense resistor, or isolation diode required
- Intelligent thermal regulation maximizes charging rate
- Intelligent recharge function
- Precharge voltage: $4.2V \pm 1\%$
- C/10 charging termination
- 2.9V trickle charge threshold
- Package: SOT23-5

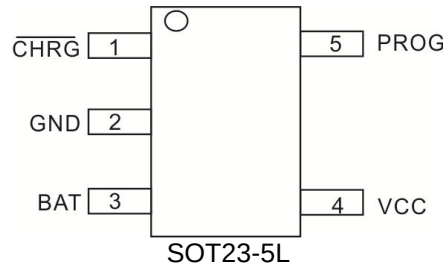
Applications

- Mobile phones, PDAs, MP3/MP4 players
- Bluetooth headsets, GPS devices
- Charging docks
- Portable devices such as digital cameras and mini speakers

Typical application circuit



Pinout



Ordering Information

Packaging	Order Model	Packaging Format	Product Marking
SOT23-5L	TP4054	Tape and Reel	TP4054

Limiting Parameters (Note 1)

Symbol	Parameter	Rated Value	Unit
VCC	Input Power Supply Voltage	-0.3~7	V
PROG	PROG Pin Voltage	-0.3~0.3	V
BAT	BAT Pin Voltage	-0.3~7	V
CHRG	CHRG Pin Voltage	-0.3~7	V
T _{BAT_SHT}	BAT Pin Short Circuit Duration	Continuous	-
I _{BAT}	BAT Pin Current	600	mA
I _{PROG}	PROG Pin Current	600	uA
T _{OP}	Operating Ambient Temperature	-40~85	°C
T _{STG}	Storage Temperature	-65~125	°C
ESD	HBM	2000	V
	MM	200	V

Note 1: Maximum limits refer to values exceeding this operating range that may damage the chip.

Electrical Parameters (Notes 2, 3)

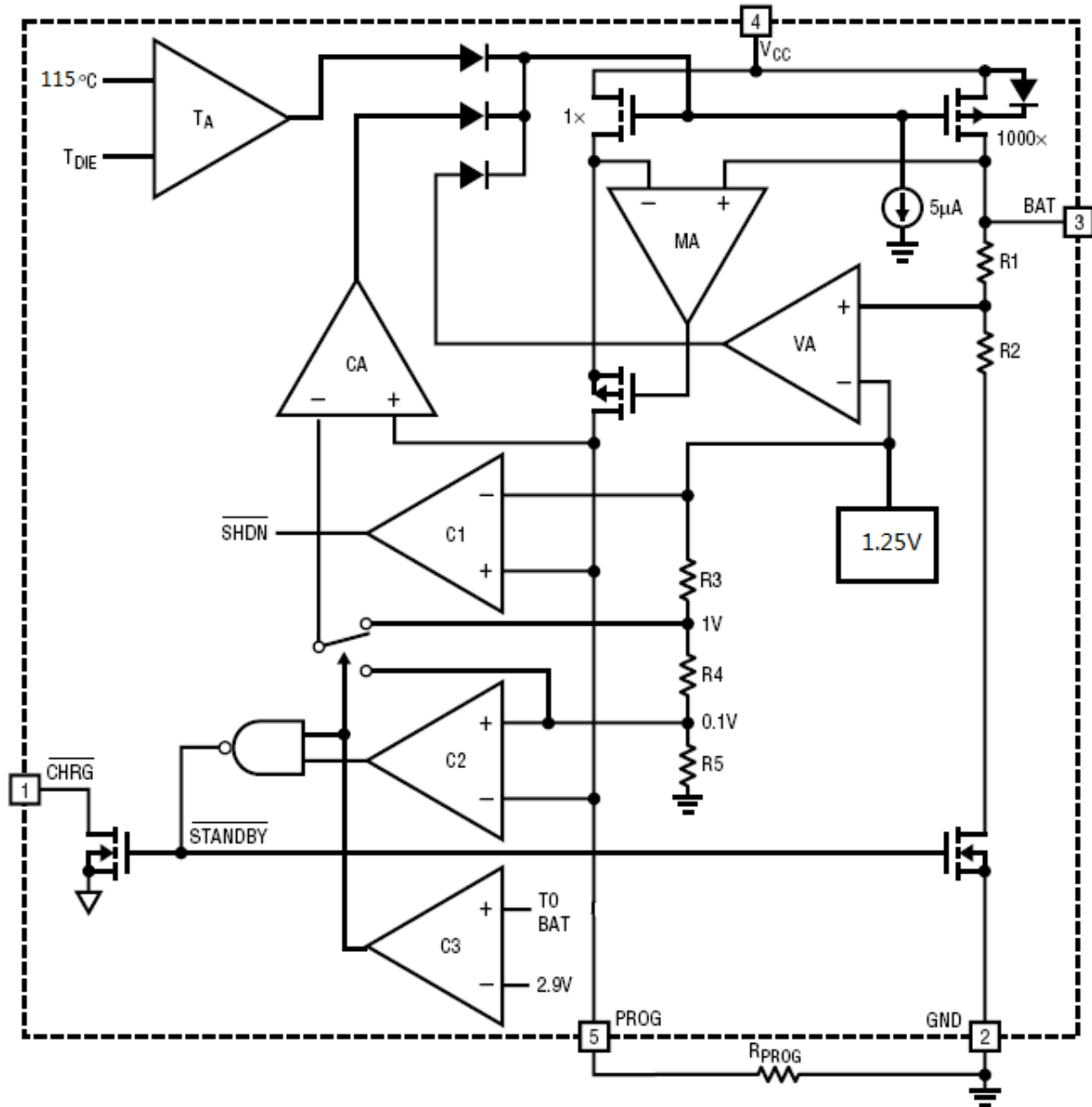
Unless otherwise specified, VIN=12V, Ta=25°C

Symbol	Parameter	Test Condition	Minimum	Typical	Maximum	Unit
V _{CC}	Input Power Supply Voltage		4.0	5	6	V
V _{FLOAT}	Output Float Charge Voltage	0°C≤T _A ≤85°C	4.158	4.2	4.242	V
I _C	Constant Current Charging Current	R _{PROG} =2K, Current Mode	450	500	550	mA
I _{TRIKL}	Trickle Charge Current	V _{BAT} <V _{TRIKL} , R _{PROG} =2K	40	60	80	mA
I _{BAT}	BAT Current Standby Mode	V _{CC} =5V, V _{BAT} =4.2V		3	5	μA
		Sleep Mode, V _{CC} =0		3	5	μA
V _{TRIKL}	Trickle Charge Threshold Voltage	R _{PROG} =2K, V _{BAT} Rise	2.8	2.9	3.0	V
V _{TRHYS}	Trickle charge hysteresis voltage	R _{PROG} =2K	60	80	100	mV
V _{UV}	V _{CC} -V _{BAT} Threshold voltage	V _{CC} rise	3.5	3.7	3.9	V
V _{UVHYS}	V _{CC} Undervoltage protection hysteresis voltage	V _{CC} fall		0.1		V
V _{ASD}	V _{CC} -V _{BAT} Threshold voltage	V _{CC} rise	60	100	140	mV
		V _{CC} fall	5	30	50	mV
		R _{PROG} =2K	40	60	80	mA
V _{PROG}	PROG Pin voltage	R _{PROG} =2K, current mode	0.9	1.0	1.1	V
V _{CHRG}	CHRG pin output low voltage	I _{CHRG} =5 mA		0.3	0.6	V
ΔV _{RECHRG}	Rechargeable battery threshold voltage	V _{FLOAT} -V _{RECHRG}	70	100	150	mV
T _{LIM}	Temperature-limited mode junction temperature			115		°C
R _{ON}	Power FET On-resistance			800		mΩ
T _{RECHRG}	Recharge comparator filter time	V _{BAT} decreases by	1	2	3	ms
T _{TERM}	End comparator filter time	I _{BAT} decreases to below I _{CHG} /10 by	1	2	3	ms

Note 2: Typical parameter values are standard parameter values measured at 25°C.

Note 3: Minimum and maximum specification ranges in the datasheet are guaranteed by testing; typical values are guaranteed by design, testing, or statistical analysis.

Internal block diagram



Operating Principle

The TP4054 is a linear charger specifically designed for single-cell lithium-ion or lithium-polymer batteries. The chip integrates a power transistor, and the charging current can be set by an external resistor. The maximum continuous charging current can reach 600mA, eliminating the need for additional choke diodes and current sensing resistors. The TP4054 includes an open-drain output status indicator to indicate whether charging is in progress or complete. During charging, the CHRG pin outputs a low level, indicating that charging is in progress; after charging is complete, the CHRG pin becomes high-impedance. If the battery voltage is below 2.9V, the TP4054 pre-charges the battery with a small current. When the battery voltage exceeds 2.9V, it uses constant current mode to charge the battery, with the charging current determined by the resistor R_{PROG} between the PROG pin and GND.

When the battery voltage approaches 4.2V, the charging current gradually decreases, and the TP4054 enters constant voltage charging mode. The charging cycle ends when the charging current decreases to the charging end threshold, which is 10% of the constant current charging current.

When the battery voltage drops below the recharge threshold of 4.1V, the TP4054 automatically begins a new charging cycle. The chip's internal high-precision voltage reference, error amplifier, and resistor divider network ensure that the accuracy of the battery-side modulation voltage is within 1%, meeting the requirements of both lithium-ion and lithium-polymer batteries. When the input voltage is below the undervoltage lockout threshold or below the battery voltage, the charger enters a low-power sleep mode, during which the battery-side current consumption is less than 3uA.

The TP4054's internal intelligent temperature control circuit automatically reduces the charging current when the chip's junction temperature exceeds 115°C. This function allows users to maximize the chip's power handling capabilities without worrying about overheating damaging the chip or external components. Thus, when designing the charging current, users do not need to consider worst-case scenarios but only typical conditions, as the TP4054 will automatically reduce the charging current in the worst-case scenario.

Pin Functions

CHRG (PIN1): Charging Status Indicator.

When the charger is charging the battery, the CHRG pin is pulled low by an internal switch, indicating that charging is in progress; when charging is finished, the CHRG pin is in a high-impedance state. If the charger is plugged in but the battery is not connected, the indicator light will flash to indicate that the battery is not connected or the battery is not making good contact.

GND (PIN2): Power ground.

BAT (PIN3): Battery positive connection terminal.

Connect the positive terminal of the battery to this pin. When there is no VCC connection or the battery is fully charged and in standby mode, the leakage current of the BAT pin is less than 3uA. The BAT pin provides charging current and a full charge voltage of 4.2V to the battery. If the battery is not connected, the floating voltage of the BAT pin is approximately 4.6V.

V_{CC} (PIN4): Positive Input Voltage.

The voltage at this pin is the operating power supply for the internal circuitry. Charging will only begin when the V_{CC} input voltage is greater than the undervoltage lockout threshold and simultaneously greater than the BAT voltage by 100mV. When the V_{CC} input voltage is lower than the undervoltage lockout threshold or the voltage difference between the V_{CC} and BAT pins is less than 30mV, the TP4054 will enter a low-power shutdown mode, in which case the current consumption at the BAT pin is less than 3uA.

PROG (PIN5): Constant Current Charging Current Setting Pin.

Connecting a resistor to GND via the PROG pin allows you to set the charging current. The setting resistor and charging current are calculated using the following formula: $R_{PROG} = 1000V / I_{BAT}$.

Determine the value of resistor R_{PROG} based on the required charging current I_{BAT}. During trickle charging, the voltage at this pin is modulated at 0.1V; during constant current charging, the voltage at this pin is fixed at 1V.

Application Notes

Charging Termination

The charging process ends when the charging current drops to 1/10 of the set value after reaching the final float charge voltage. This condition is detected by monitoring the PROG pin using an internal filter comparator. Charging terminates when the PROG pin voltage drops below 100mV for more than 2ms.

Intelligent Recharge

In standby mode, the TP4054 monitors the BAT pin voltage. A new charging cycle begins only when the BAT pin voltage drops below the recharge threshold voltage of 4.1V (corresponding to 80%–90% battery capacity). This avoids unnecessary repeated charging and effectively extends battery life.

Adding a thermal regulation resistor

Lowering the voltage drop across VCC and BAT of the IC can significantly reduce power consumption within the IC. During thermal regulation, this increases the charging current. This can be achieved by connecting a 0.5Ω resistor or a diode with a forward voltage drop less than 0.5V in series between the input power supply and VCC, thereby dissipating some power.

Soft-start charging current

The TP4054 has a built-in soft-start circuit. When a charging cycle is initiated, the charging current will gradually increase from zero to a constant current charging current over a period of 20μs.

Charging Status Indicator

CHRG is the open-drain status indicator output. When the charger is charging, CHRG is pulled low. After charging is complete, CHRG is in a high-impedance state.

If no battery is connected, the voltage at pin BAT will be approximately 4.6V when the charger is plugged in, and the indicator light will flash to indicate that no battery is connected or the battery connection is poor. If the status indicator function is not used, CHRG should be left floating or grounded.

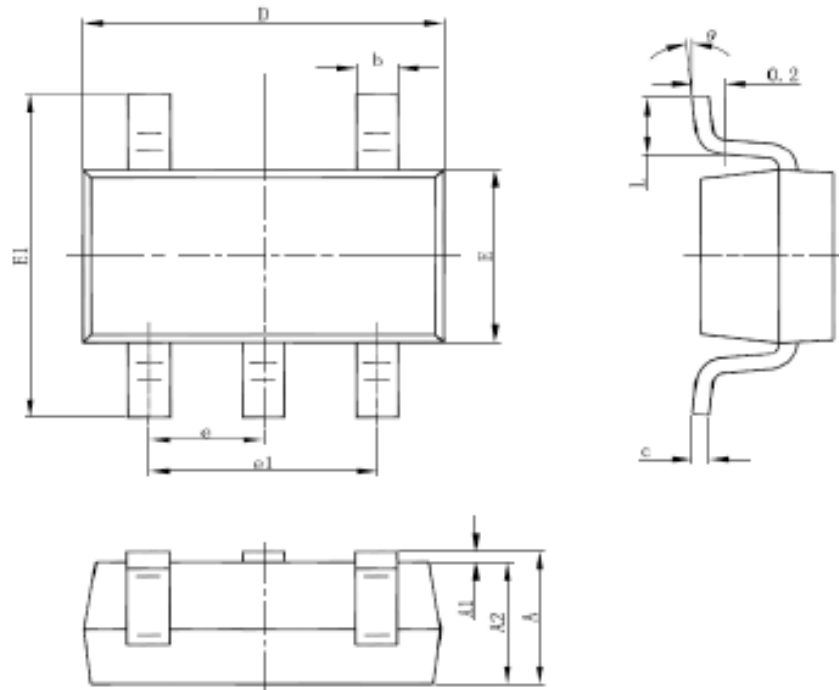
Intelligent Temperature Control

The TP4054 integrates intelligent temperature control. When the chip temperature exceeds 115°C, it automatically reduces the charging current. This function allows users to increase the upper limit of the power handling capacity of a given board without risking damage to the TP4054. The charging current can be set based on typical (not worst-case) ambient temperature, ensuring that the charger automatically reduces the current under worst-case conditions.

Package Outline Dimensions

SOT23-5L

SOT-23-5L PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E	1.500	1.700	0.059	0.067
E1	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
θ	0°	8°	0°	8°